

FINAL EXAM

ECON301—ECONOMETRICS I

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Instructions. The exam has a total of 100 points. Read the problems carefully; then decide which ones you want to attempt first. I suggest that you try to answer first the problems which you feel most confident about and then move on to the ones in which you don't feel so confident. **Write your answers neatly and be organized; cross anything that you don't want me to consider. Please number each page you use.** The duration of the exam is 150 minutes plus 30 minutes of submission time (total of 180 minutes or 3 hours).

1. (25 points.) Let Y denote a Bernoulli(θ) random variable with $0 < \theta < 1$. Suppose you have access to a random sample $\{Y_i\}_{i=1}^n$ from Y . Let $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n Y_i$.

- (a) (2 points.) Show that $\hat{\theta}_n$ is a consistent estimator of θ .
(b) (2 points.) Show that $\sqrt{n}(\hat{\theta}_n - \theta) \rightarrow_d N(0, \theta(1 - \theta))$.
(c) (2 points.) Suppose we are interested in the *odds ratio*, $\gamma = \theta/(1 - \theta)$, which is the probability of success over the probability of failure. Propose a consistent estimator of γ and show that it is indeed consistent. Call your proposed estimator $\hat{\gamma}_n$.
(d) (5 points.) Consider the following theorem:

Theorem (Delta Method). If $\sqrt{n}(T_n - \tau) \rightarrow_d N(0, \psi^2)$ and $U_n = h(T_n)$ where h is continuously differentiable at τ , then

$$\sqrt{n}(U_n - h(\tau)) \rightarrow_d N(0, [h'(\tau)]^2 \psi^2).$$

Equivalently, if $T_n \sim AN(\tau, \psi^2/n)$ and $U_n = h(T_n)$ is continuously differentiable at τ , then

$$U_n \sim AN\left(h(\tau), \frac{[h'(\tau)]^2 \psi^2}{n}\right).$$

Use the Delta Method to show that $\sqrt{n}(\hat{\gamma}_n - \gamma) \rightarrow_d N(0, \theta/(1 - \theta)^3)$.

- (e) (4 points.) Propose a consistent estimator of $\theta/(1 - \theta)^3$ and show that it is indeed consistent. Call your proposed estimator s_n^2 .
(f) (5 points.) Take the result in (d) and the existence of the consistent estimator in (e) that I asked you to show as facts. Give the details of building an asymptotic pivot to make inference on γ .
(g) (5 points.) Use your result in (f) to construct an asymptotic $(1 - \alpha)\%$ confidence interval for γ .
2. (25 points.) Consider the following best linear predictor

$$E^*[Y|X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 X_2. \quad (1)$$

and let $\hat{\beta}$ be the LSE estimator of model (1). Now consider the following alternative best linear predictor

$$E^*[Y|X_1^*, X_2] = \delta_0 + \delta_1 X_1^* + \delta_2 X_2, \quad (2)$$

where $X_1^* = cX_1$ and c is a constant. Let $\hat{\delta}$ be the LSE of model (2). What is the relationship between $\hat{\beta}$ and $\hat{\delta}$?

3. (25 points.) Using a sample of recent high school graduates in the US, you are interested in estimating the following BLP:

$$E^*[\text{sat}|\text{hsize}, \text{hsize}^2, \text{female}, \text{black}, \text{female} \times \text{black}] = \beta_0 + \beta_1 \text{hsize} + \beta_2 \text{hsize}^2 + \beta_3 \text{female} + \beta_4 \text{black} + \beta_5 \text{female} \times \text{black} \quad (3)$$

where the variables are: *sat* is the combined SAT score; *hsize* is the size of the student's high school graduating class, in hundreds; *female* is a gender dummy variable; *black* is a race dummy variable equal to one for blacks, and zero otherwise.

The estimation results are in Table 1. The standard errors are in parenthesis, below the estimated coefficient. The estimate for the variance covariance matrix is provided in table 2.

- (a) (4 points.) Is there strong evidence that $hsize^2$ should be included in the model? From this equation, what is the optimal high school size?
- (b) (4 points.) Holding $hsize$ fixed, what is the estimated difference in SAT score between nonblack females and nonblack males? Construct a hypothesis test in which the null hypothesis is that there is no difference between nonblack females and nonblack males vs the alternative that there is a difference. Test it at the 1% level.
- (c) (4 points.) What is the estimated difference in SAT score between nonblack males and black males? Test the null hypothesis that there is no difference between their scores, against the alternative that there is a difference at the 5% level.
- (d) (6 points.) What is the estimated difference in SAT score between black females and nonblack females? Construct the relevant hypothesis test to test whether this difference is significant or not and test it at the 1% level.
- (e) (7 points.) Consider student A , which graduated from a high school class of 200 ($hsize = 2$), is *female and black*. Student B graduated from a high school class of 300 ($hsize = 3$) and is also *female and not black*. Let $\Delta sat = sat_B - sat_A$ denote the difference in SAT scores where sat_B and sat_A are the SAT scores of students B and A respectively. Delineate a procedure that will allow you to construct a 95% confidence interval around Δsat . Use your proposed procedure to estimate the confidence interval.

Table 1: Estimation results from the model (3)

Dependent variable:	
	sat
$hsize$	19.297*** (3.814)
$hsizesq$	-2.195*** (0.524)
$female$	-45.091*** (4.226)
$black$	-169.813*** (15.423)
$female \times black$	62.306*** (19.305)
Constant	1,028.097*** (6.355)
Observations	4,137
R^2	0.086
Adjusted R^2	0.085
Note: *p<0.1; **p<0.05; ***p<0.01	

Table 2: Estimated Variance-Covariance Matrix

$$\hat{\Omega} = \begin{bmatrix} 40.386516 & -20.199194 & 2.335095 & -9.031893 & -11.536296 & 8.210525 \\ -20.199194 & 14.545411 & -1.900289 & 0.078404 & 1.261833 & 0.068763 \\ 2.335095 & -1.900289 & 0.274387 & -0.000038 & -0.077895 & 0.031187 \\ -9.031893 & 0.078404 & -0.000038 & 17.863001 & 8.820194 & -17.816754 \\ -11.536296 & 1.261833 & -0.077895 & 8.820194 & 237.863505 & -237.367652 \\ 8.210525 & 0.068763 & 0.031187 & -17.816754 & -237.367652 & 372.702161 \end{bmatrix}$$

4. (25 points.) Consider the following house pricing equation

$$E^*[price|assess] = \beta_0 + \beta_1 assess, \quad (4)$$

where *price* is the house final selling price; *assess* is the assessed housing value (before the house was sold). The assessed value is usually calculated by a real estate professional.

We say that the assessment is *rational* if $\beta_1 = 1$ and $\beta_0 = 0$. Equation (4) was estimated using data collected from the real estate pages of the Boston Globe in 1990, using information about house selling in Boston, MA. The results from this estimation procedure can be found in the column titled (4) in table 3 and the estimated variance-covariance matrix in table 4.

- (a) (4 points.) First, test the hypothesis that $H_0 : \beta_0 = 0$ against the two-sided alternative. Then, test $H_0 : \beta_1 = 1$ against the two-sided alternative. What do you conclude?
- (b) (7 points.) Now, test the joint hypothesis that $\beta_0 = 0, \beta_1 = 1$. versus the two-sided alternative that either $\beta_0 \neq 0$ or $\beta_1 \neq 1$.
- (c) (7 points.) Compare the results from (a) and (b). Which test would you use to assess rationality of the assessment?
- (d) (7 points.) Now consider the following model:

$$E^*[price|assess, lotsize, sqrft, bdrms] = \beta_0 + \beta_1 assess + \beta_2 lotsize + \beta_3 sqrft + \beta_4 bdrms, \quad (5)$$

where *lotsize* is the size of the lot in square feet; *sqrft* is the size of the house in square feet; and *bdrms* is the number of bedrooms. The estimation results are in the column titled (5) in table 3 and the estimated variance-covariance matrix in table 5. Use that information to test $H_0 : \beta_2 = \beta_3 = \beta_4 = 0$ at the 5% significance level.

Table 3: Estimation results from the house pricing models.

	Dependent variable:	
	price	
	(4)	(5)
assess	0.976*** (0.067)	0.908*** (0.116)
lotsize		0.001*** (0.0002)
sqrft		-0.001 (0.017)
bdrms		11.602** (5.392)
Constant	-14.472 (20.382)	-38.887* (23.082)
Observations	88	88
R ²	0.820	0.829
Adjusted R ²	0.817	0.821
Note:	*p<0.1; **p<0.05; ***p<0.01	

$$\hat{\Omega} = \begin{bmatrix} 415.409273 & -1.33684 \\ -1.33684 & 0.004516 \end{bmatrix}$$

Table 4: Estimated Variance-Covariance Matrix for model (4)

$$\hat{\Omega} = \begin{bmatrix} 532.770615 & -0.7128 & -0.000586 & -0.043468 & -61.894163 \\ -0.7128 & 0.013423 & -0.000006 & -0.001621 & -0.017566 \\ -0.000586 & -0.000006 & 0 & 0.000001 & -0.000275 \\ -0.043468 & -0.001621 & 0.000001 & 0.000288 & -0.014739 \\ -61.894163 & -0.017566 & -0.000275 & -0.014739 & 29.076399 \end{bmatrix}$$

Table 5: Estimated Variance-Covariance Matrix for model (5)